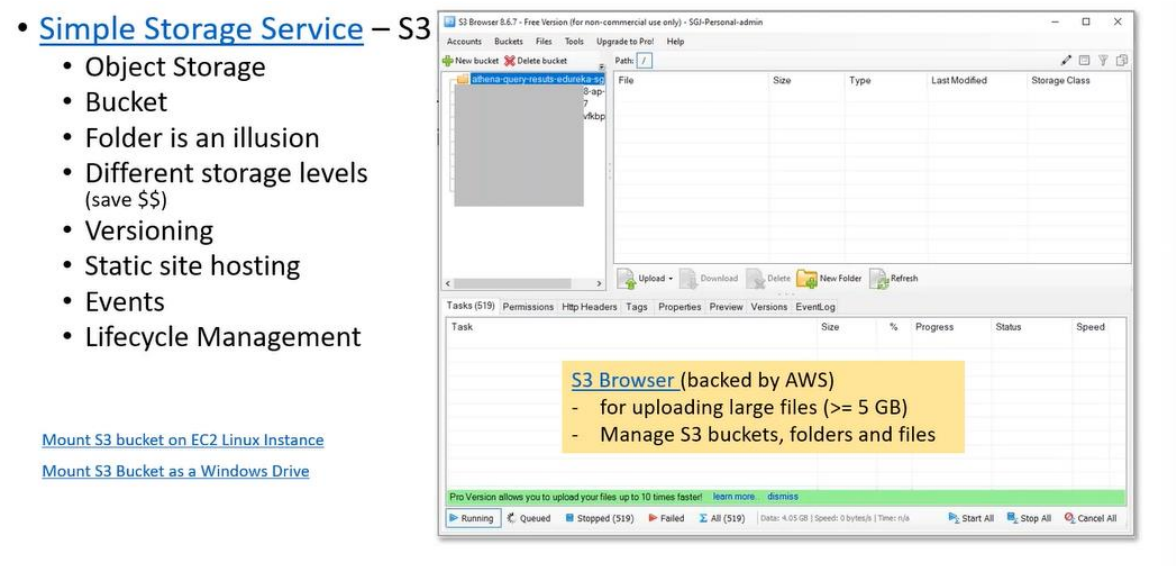
Object Storage- Client (S3 browser)

Encrypt the data, transfer the data through public internet.

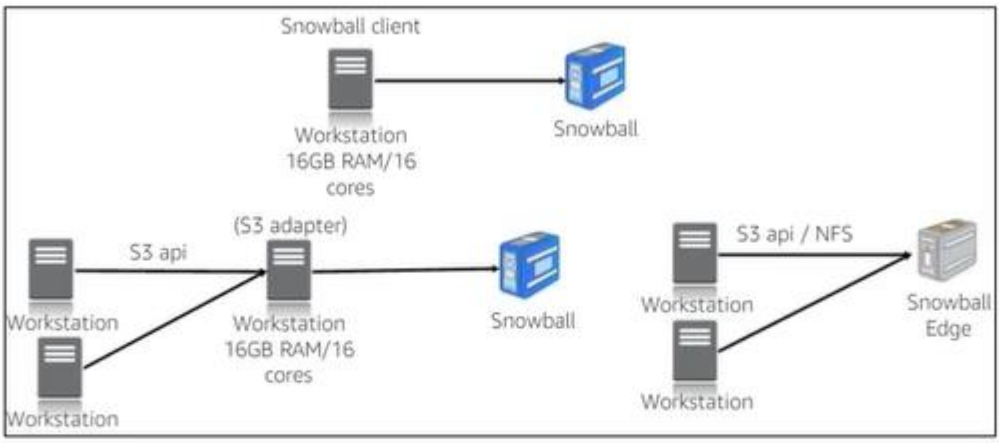


mount S3 bucket on EC2 instance (TNT Drive).

Object Storage- Client (WinSCP)

Using the script, automate the data transfer.

Snow Family



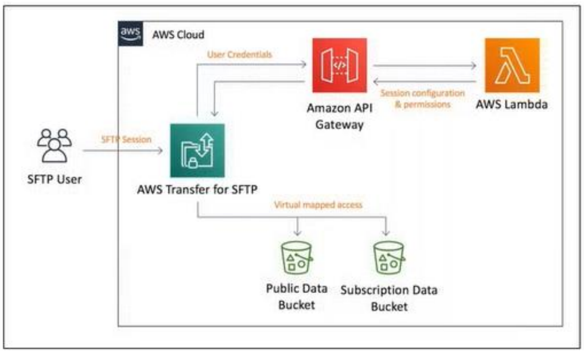
Helps eliminate challenges that can be encountered with large-scale data transfers including high network costs, long transfer time and security concerns.

Use for import data to cloud or export data from cloud at computing locations.

- Snowcone
- Snowball Edge

- Snow Mobile

AWS Transfer Family (SFTP,FTPS,FTP)



Benefits
Reduce operational burdens
Streamline workflow migrations
Integrate processing and management
API can be enabled on service for programmatic file transfers
If S3 is used to host static web site, this transfer service can help you manage contents without exposing AWS console to the third party or resource involved

- SFTP, FTPS, FTP (exchange data on cloud)
- SFTP: SSL (end point ssl)
- FTPS: TLS
- FTP: non secure

step1: create a server.

step2: select the protocol (SFTP).

step3: Choose an Identity provider (service managed).

step4: Choose an end point

step5: Choose a domain

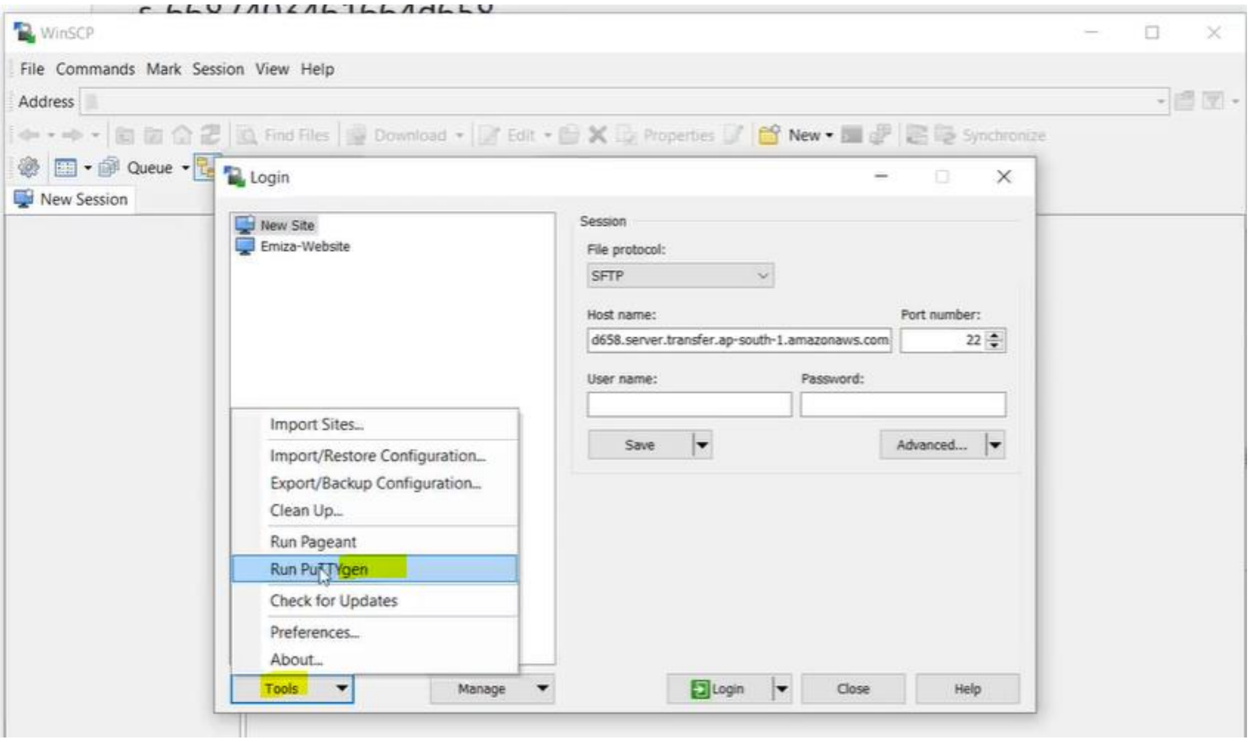
step6: Configure additional details

How app securely communicate?

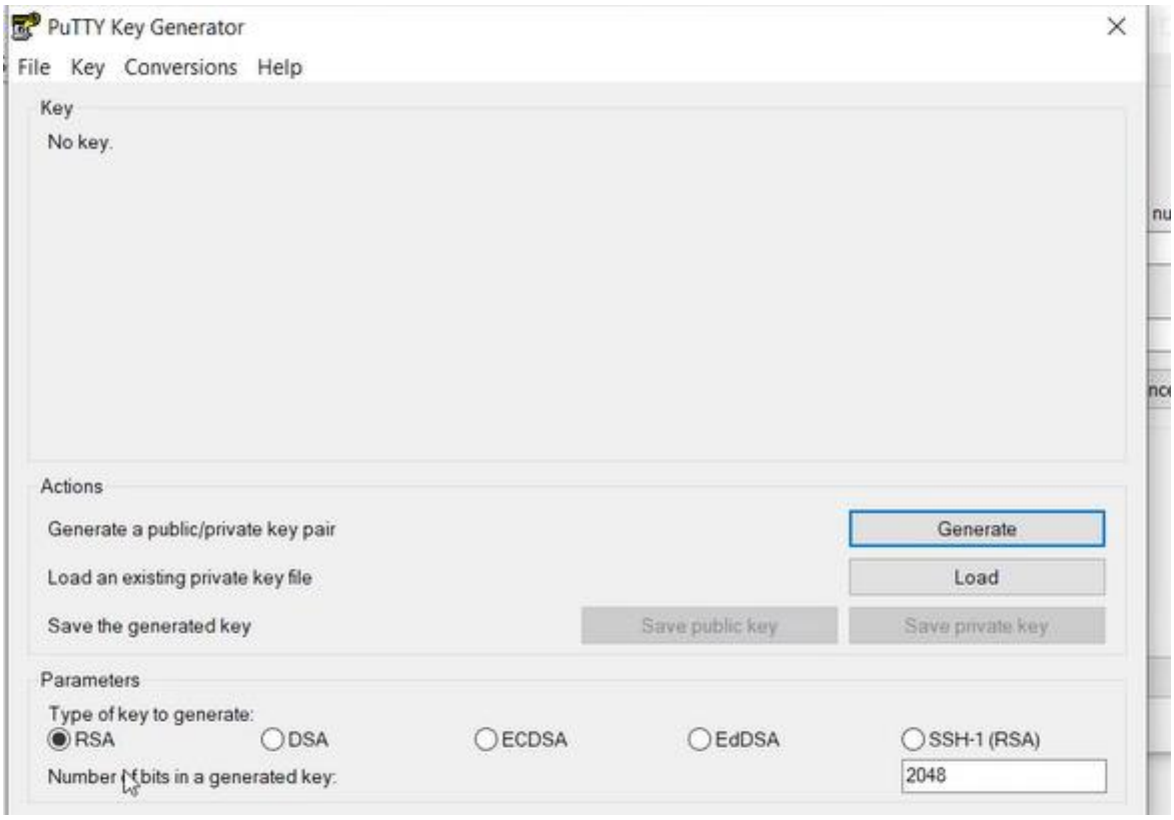
Concept: Encrypt Data using the public key can be only decrypted using corresponding private key.

step7: Connect SFTP server through WinSCP

We have to create a user and every user carry a certificate, certificate is a public/private key generated through the putty key generator



- click on generate button (move key randomly to generate key)



- save public key (RSA 2048)
- save private key (RSA 2048)

step8: click on SFTP server

- Add User
    - step8.1: create a role for Transfer (S3Full Access)
    - User Name
    - IAM role – sftp role
    - Home Directory ? S3 bucket (dump the data)
    - SSH public key- copy the public key
    - Click on Add button

WinSCP:

connect to the SFTP server.

- host: SFTP end point
- user name:
- password: (Advanced)
- password-> authenticate-> private key
- click on login

copy the files, available is S3 bucket.

How to retain timestamp?

Tools-> preferences->transfer-> default-> edit (uncheck set permissin,preserve time stamp)

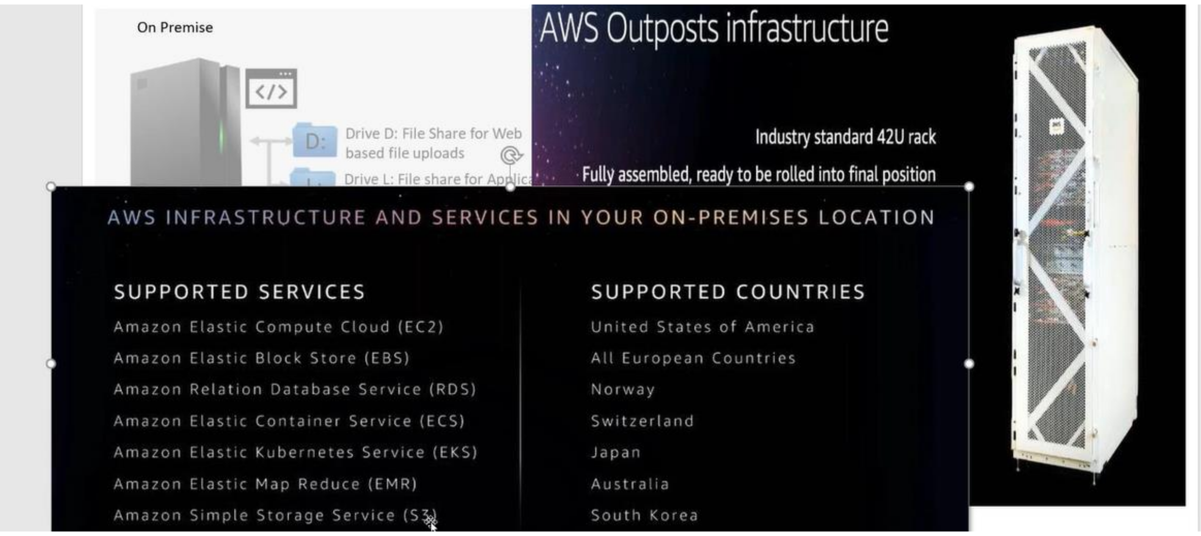
Diff between DataSync and S3 Tranfer Acceleration?

- DataSync provides two way communication, can switch the endpoints
- DatSync also provides validation feature.
- S3 Transfer Acceleration also don't ensure about data inegrity.

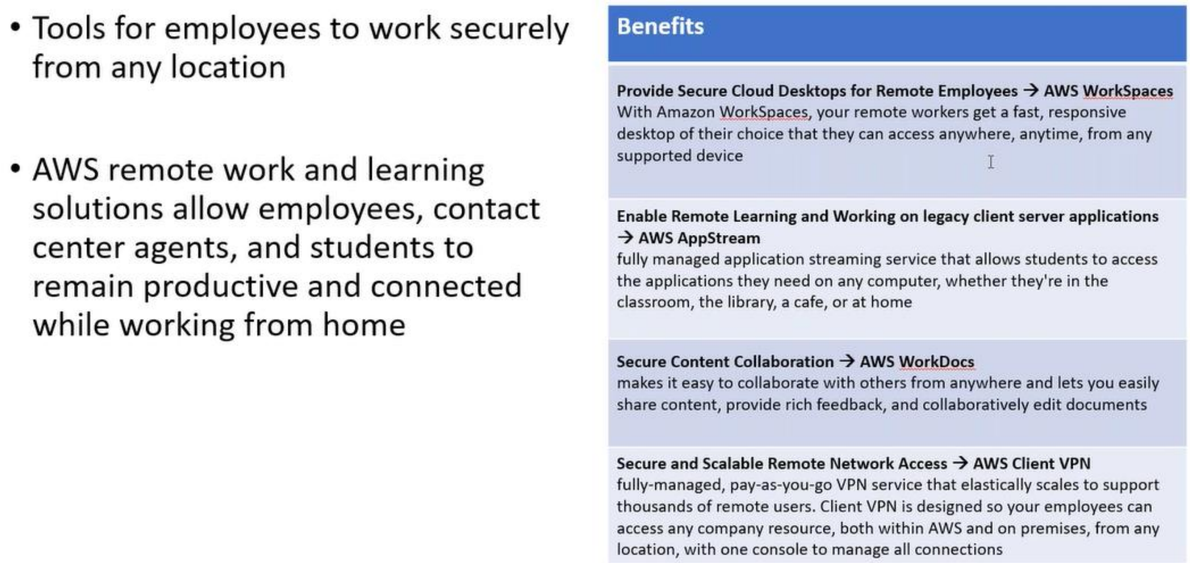
UC:

- Direct Connect is so costly $.
- Not sure if we can operate in cloud.
  - Need to explore
- Not comfortable transferring data on the public internet.
- What would be the net benefit post migration?

Outposts:

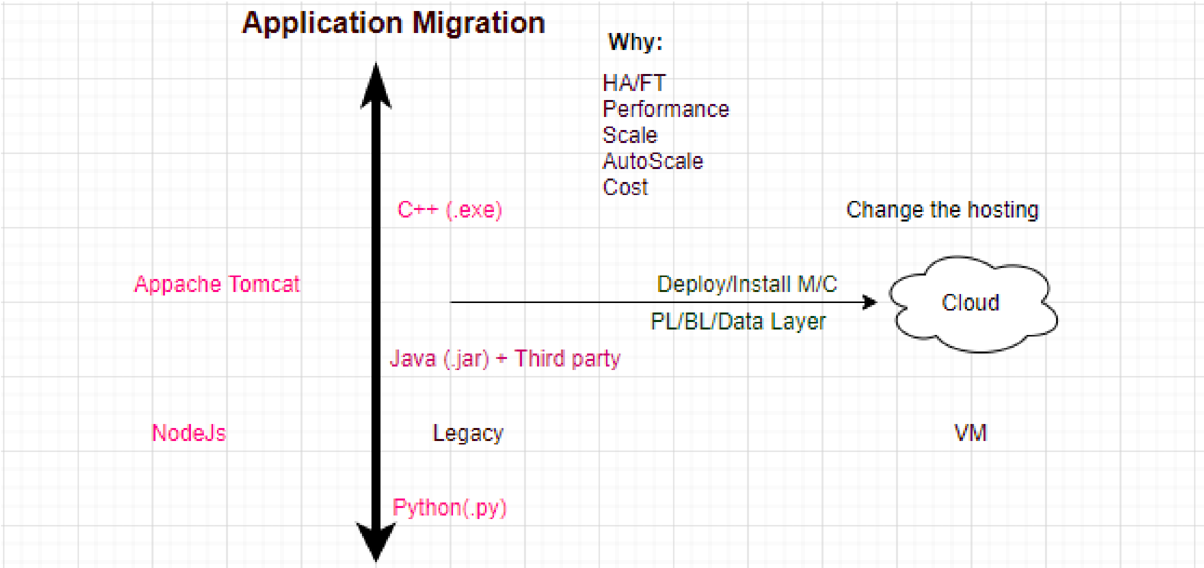


AWS remote work and learning:

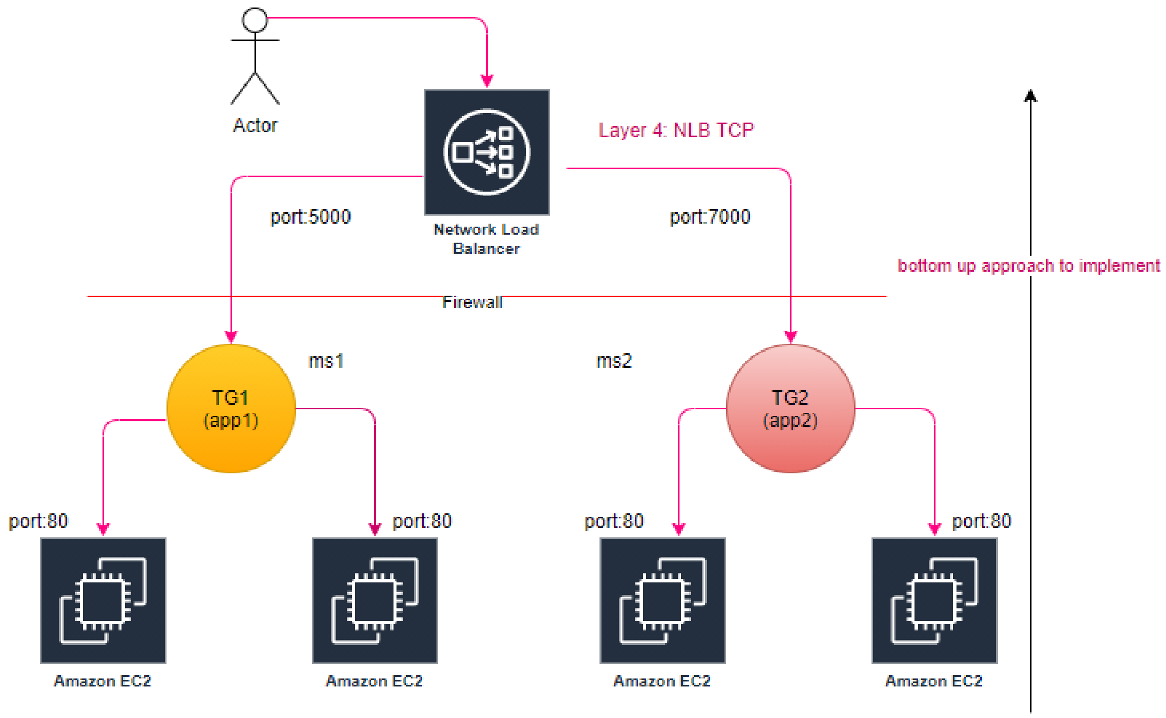


Application Migration:

What is application migration?

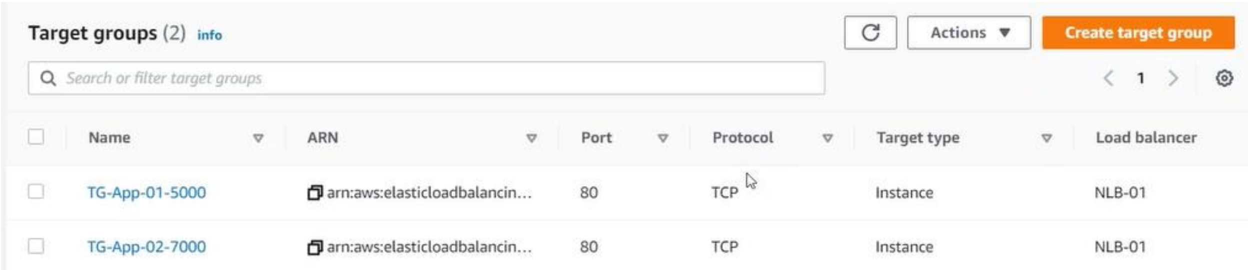


ELB (Elastic Load Balancing): NLB uses burst algo to balance the traffic.



step1: Launch 2-2 instances per AZ

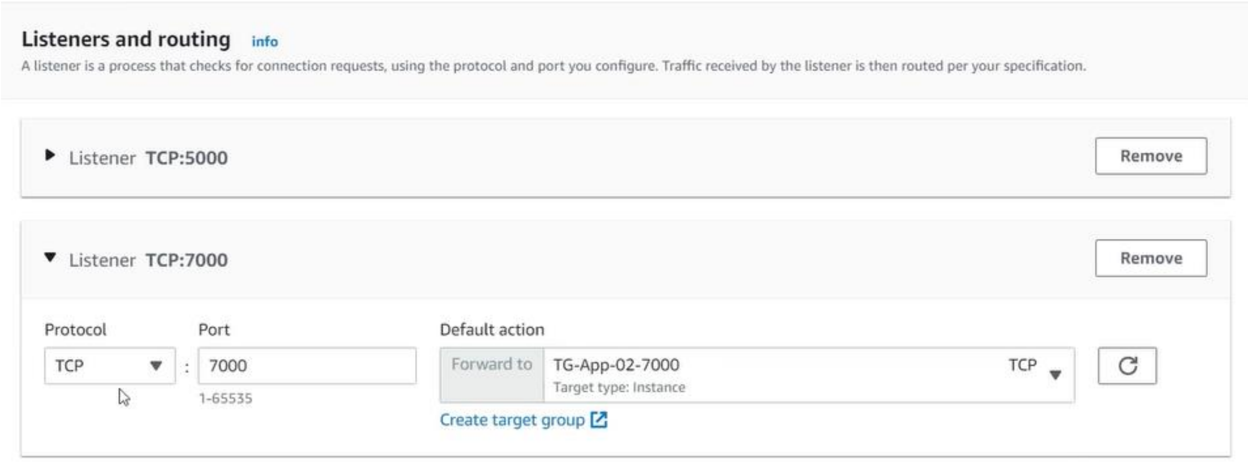
step2: Create TG for NLB (TCP), actual load balancing happens at TG level, **health check always be HTTP**



step3: Register the Instaces (targets) with TG.

step4: Configure Routing→ Configure Target Group

Mapping-> corresponding to each subnet(routing) a seprate NIC is created.



These ports are immediately dropped as traffic routed

Validate:

- Is NLB end point is active.
- Is target group's targets are healthy
- This is also called reverse proxy?

You can host multiple applications behind single end point and redirect the traffic based on the port number.

Tab: Integrated Services

- VPC Endpoint Service (AWSPrivateLink) Layer4
  - You can create your NLB as a service, for others to consume as a VPC Endpoint Service.
- Traffic Mirroring (Layer4)
- Global Configuration (pop)
- Config (record each activity in the account)

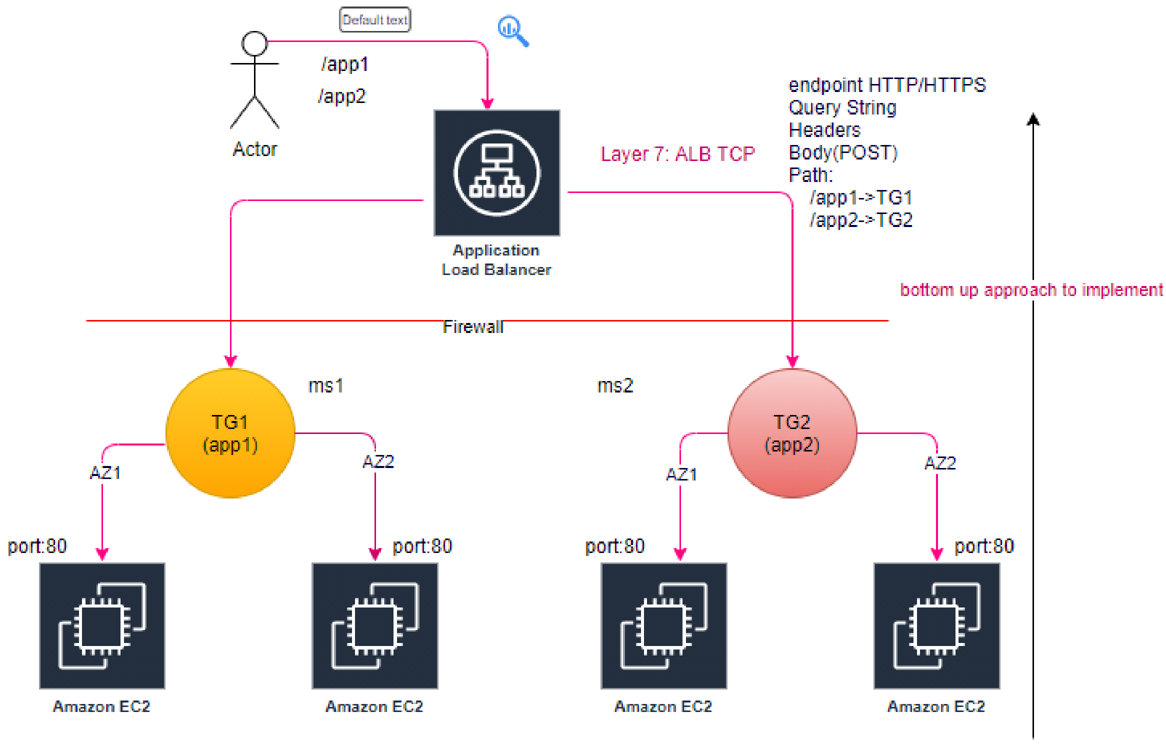
Endpoint: From consumer point of view

Endpoint Service-> publisher/provider

As a vpc end point can access within the VPC, while using endpoint service other can access.

All applications worked at region labels.

ELB (Elastic Load Balancing): ALB, uses round robin algo to balance the traffic.



step1: Launch 2-2 instances per AZ

step2: Create TG for NLB (TCP), actual load balancing happens at TG level, **health check always be HTTP**

Target groups (1/3) info

Actions ▼ | Create target group

Search or filter target groups

Name	ARN	Port	Protocol	Target type	Load balancer
TG-01	arn:aws:elasticloadbalancin...	80	HTTP	Instance	-
TG-App-01-5000	arn:aws:elasticloadbalancin...	80	TCP	Instance	NLB-01
TG-App-02-7000	arn:aws:elasticloadbalancin...	80	TCP	Instance	NLB-01

step3: Register the targets with TG.

step4: Configure Routing→ Configure Target Group

Mapping-> corresponding to each subnet(routing) a seprate NIC is created.

Validate:

- Is ALB end point is active.
- Is target group’s targets are healthy
- This is also called reverse proxy?

You can host multiple applications behind single end point and redirect the traffic based on the http/https, host-based routing (IP), path-based routing, http header-based, http-method based, Query String parameter-based, Source IP address CIDR based.

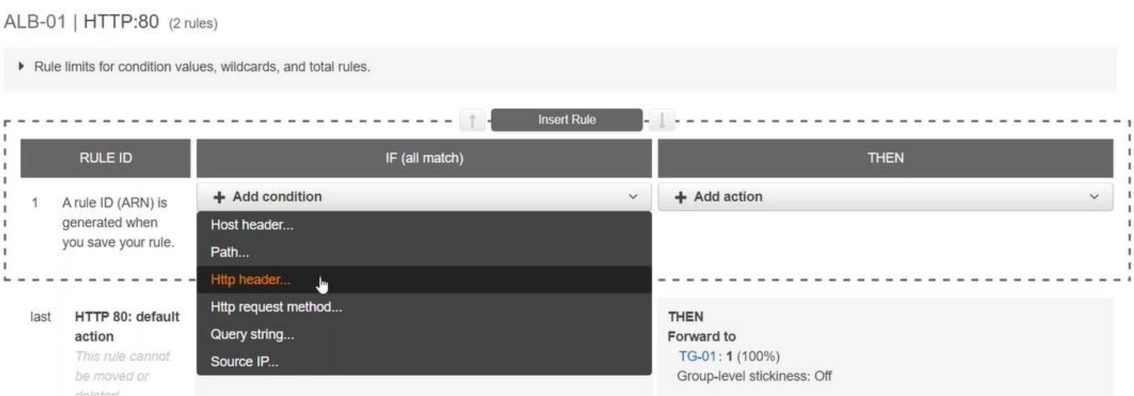
Tab: Integrated Services

- AWS WAF (Layer7)
- Global Configuration (pop)
- Config (record each activity in the account)

How to set Query based load balancing?

Tab: Listener

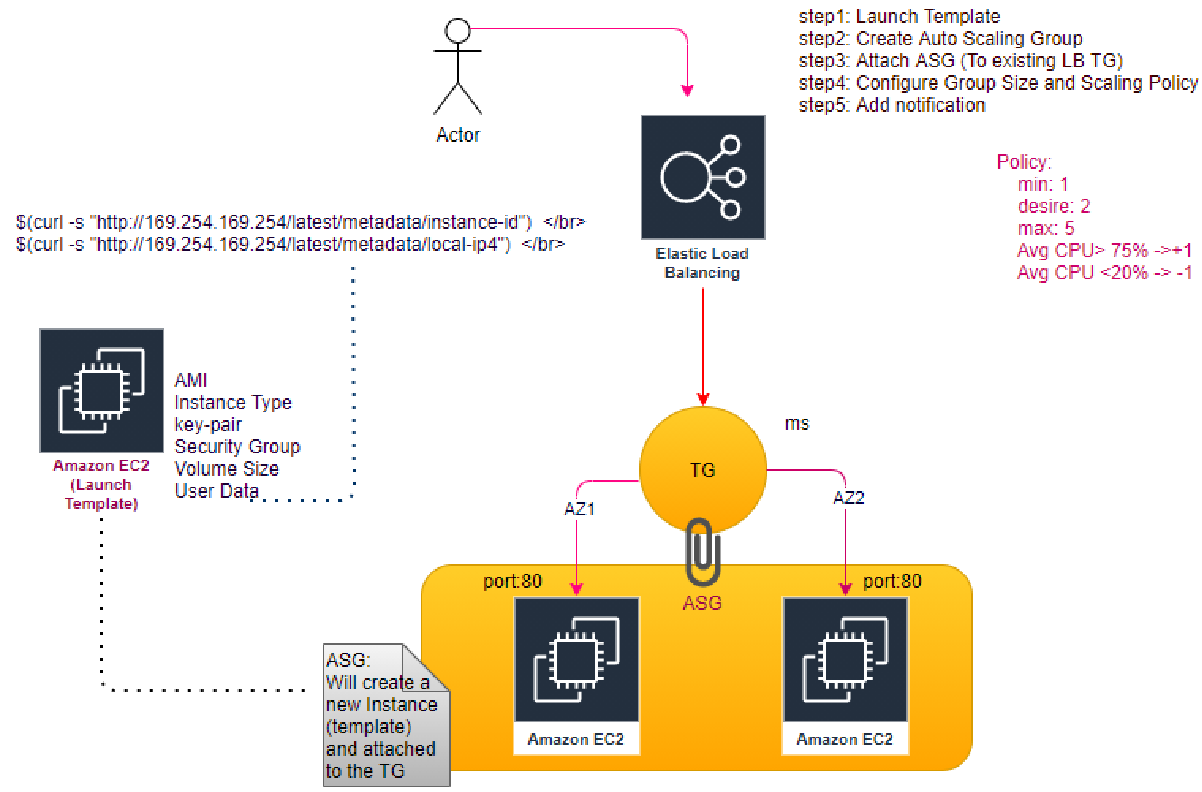
step1: Click on view/edit rule



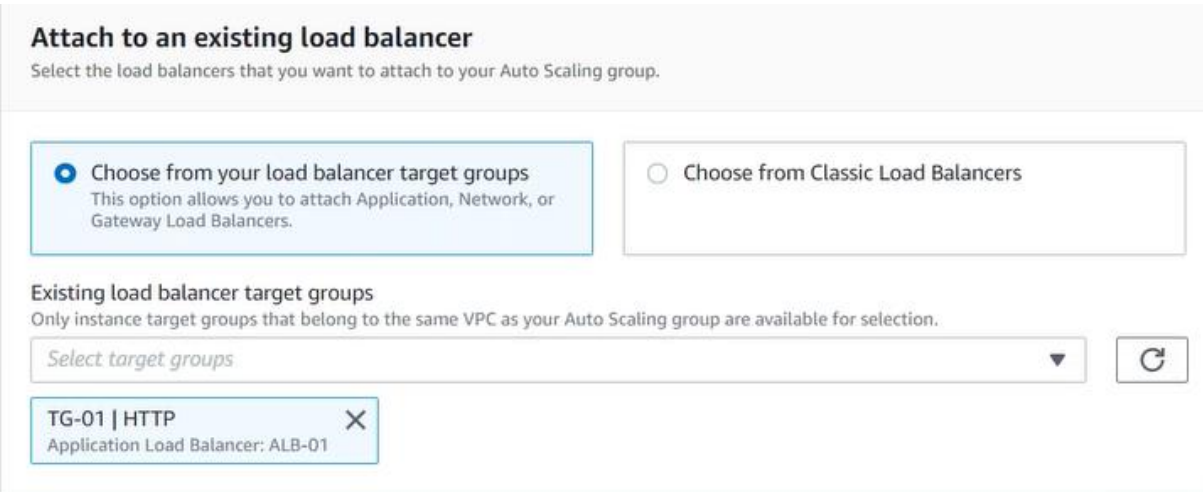
#	Routing Option	Route Client Request based on
1	Host-based	HOST field in HTTP header
2	Path-based	URL path on HTTP header
3	HTTP header-based	Value of any standard field or custom HTTP header
4	HTTP method-based	Any standard or custom HTTP method
5	Query String parameter-based	Query String or Query Parameter
6	Source IP address CIDR-based	Source IP address CIDR from where the request originates

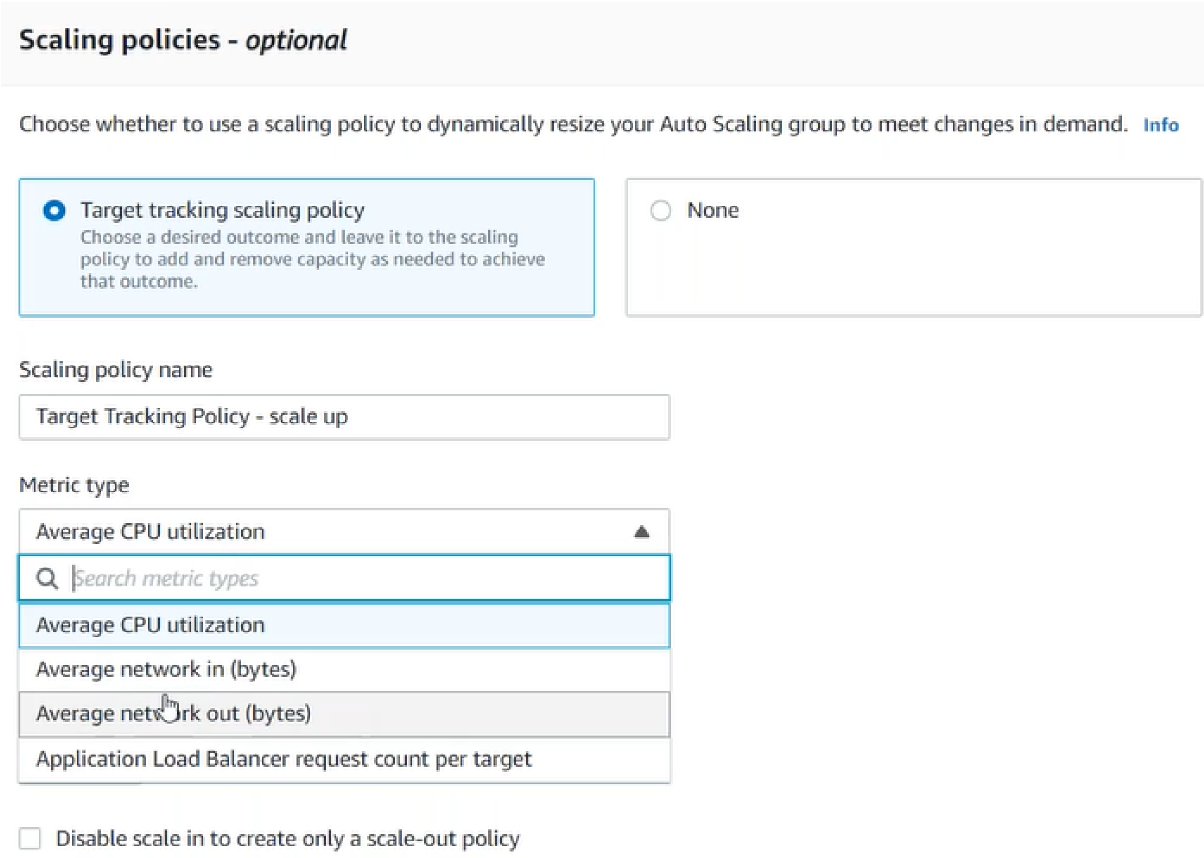
All applications worked at region labels.

Autoscaling (HPA):



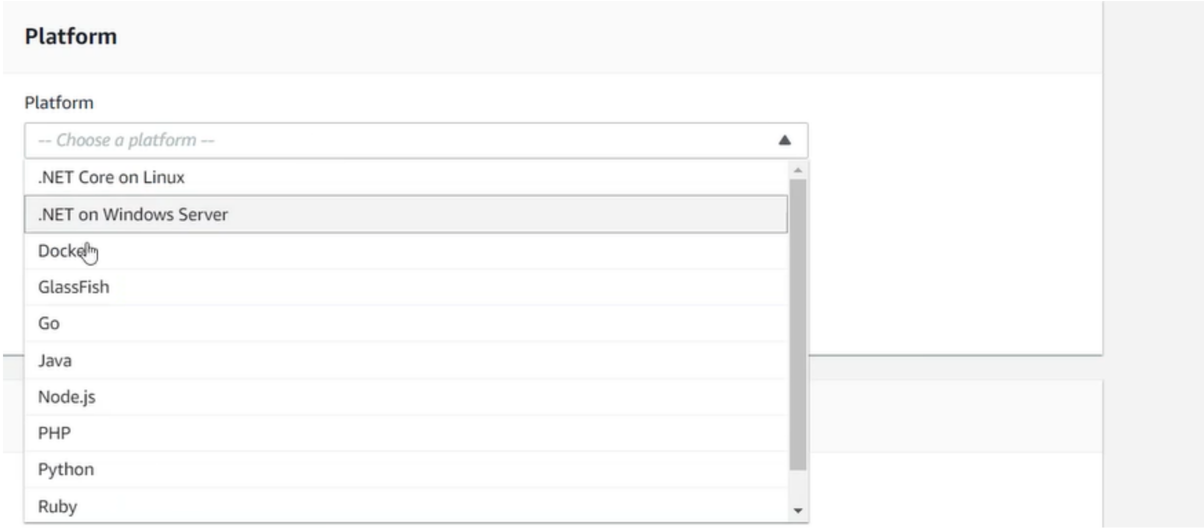
step: Just map Auto Scaling group with the target group, along with policy, will automatically creates EC2 instance as per policy.



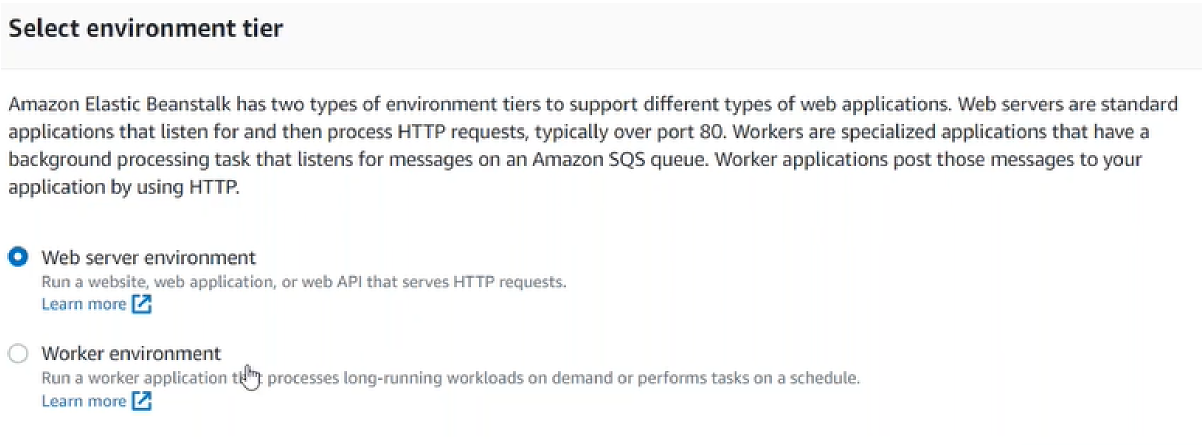


AWS Beanstalk (PaaS): Just handover your application to it (CloudFormation script executed internally).

- Auto Scaling
- Load Balancing
- Monitoring
- Logging
- Tracking
- Deployment Options
- Application Platforms
- Customization
- Web, Batch, Containers



Using Beanstalk we can migrate our application, using the environments



It has its own limitation.

CloudFormation (IaaS): Stack to create AWS resources, Beanstalk creates a default (readymade) stack for given application, while using CloudFormation we can create customized (as per our requirement) stack.